

VITECH ENVIRO SYSTEMS

AI ENGINEERING KNOWLEDGE BASE

Paint Booth Selection, Design Calculations, Product Database & Engineering Rules

Prepared as a structured technical input document for an AI engineering / proposal / quotation platform. This document defines the data model, engineering workflow, calculation framework, selection logic, BOM structure and validation rules that should be encoded into the AI.

Important: VITECH-specific constants, standard sizes, material thicknesses, filter ratings, allowable velocities, pressure-drop limits, safety factors, equipment makes/models and commercial rates must be replaced or confirmed from approved VITECH drawings, calculations, specifications and historical projects. Generic formulas in this document are engineering framework, not authorization to invent VITECH design values.

1. AI PRODUCT DATABASE – MASTER STRUCTURE

Required hierarchy

Customer Requirement → Product Family → Type → Model/Configuration → Design Inputs → Calculations → Selection → Components → BOM → Costing → GA/P&ID; → Technical Specification → Validation → Quotation

Required fields for every VITECH product

- Product ID, family, category, type, subtype, model/configuration, application, industry and process.
- Capacity range, standard/custom status, dimensions, material of construction, thickness/grade and construction method.
- Design inputs, calculation IDs, formulas, outputs, selection rules, component IDs, BOM ID, GA drawing ID, P&ID; ID and specification ID.
- Applicable standards, assumptions, safety factors, validation rules, engineering notes, source document and revision.
- Historical project reference, quotation reference, customer requirement, approval status and engineer/date/revision.

2. VITECH PRODUCT FAMILIES TO STRUCTURE

Family	Main types / variants
Paint Booth	Cross Draft, Downdraft, Side Draft, Open Front, Closed, Dry Filter, Wet/Water Curtain, Custom
Powder Coating	Manual, Automatic, Batch, Conveyorized, Cyclone Recovery, Cartridge Recovery
Pretreatment	Dip, Spray, Degreasing, Derusting, Phosphating, Chromating, Water Rinse, Dry-off
Ovens	Batch, Conveyorized, Dry-off, Curing, LPG/Gas, Diesel, Electrical
Dust Collection	Cyclone, Bag Filter, Cartridge Filter, Unit Dust Collector, Pulse Jet
Scrubbers	Spray Tower, Packed Bed, Orifice, Horizontal Wet, Dry/Activated Carbon
Fume Extraction	Tank Extraction, Paint Fume, Process Extraction, Central/Local Exhaust
Material Handling	Overhead, Monorail, Floor, Trolley, Power & Free, Automatic
Clean/Controlled Systems	Clean Room and process-controlled environments
Special Systems	Chip & Dust Collection, Forced Air Cooling, Pneumatic Conveying Dryer

3. PAINT BOOTH – REQUIRED CUSTOMER INPUTS

- Booth/application: wet painting, powder coating, drying, touch-up, inspection or special process.
- Product/component length, width, height and maximum weight.
- Required internal booth length, width and height or available floor/headroom.
- Paint type, coating process, paint consumption, transfer method, manual/automatic operation and number of operators/guns.
- Production rate: parts/hour, batch size, operating hours/day and required cycle time.

- Required finish quality and overspray characteristics.
- Indoor/outdoor installation, ambient conditions and available electrical supply.
- Entry/exit arrangement, doors, product loading method and conveyor interface.
- Required airflow direction: cross draft, downdraft, side draft or other approved arrangement.
- Required emission/filtration standard and exhaust arrangement.

4. HOW TO CHOOSE THE PAINT BOOTH

Step 1 – Identify the process

- If the process is wet liquid paint → select a wet-paint booth family.
- If the process is powder coating → select powder coating booth/recovery family.
- If the requirement is mainly drying/curing → evaluate oven/drying equipment rather than a paint booth.
- If overspray loading is high or wet capture is required → evaluate wet/water-wash configuration.
- If dry filtration is acceptable → evaluate dry-filter booth.

Step 2 – Identify airflow configuration

- Cross Draft: airflow travels generally from inlet/filter side toward exhaust/filter side; suitable when layout and process permit horizontal flow.
- Downdraft: air moves from top toward floor extraction; suitable where floor extraction and appropriate booth/floor filtration are feasible.
- Side Draft: air is extracted through side/rear arrangement; useful where floor extraction is impractical.
- Open Front: use when the process and risk assessment permit an open working face and the required capture airflow can be maintained.
- Closed Booth: use when enclosure/containment and controlled airflow are required.

Step 3 – Determine design dimensions

Internal Booth Length $\geq$ Product Length + Required Working/Access Clearance

Internal Booth Width $\geq$ Product Width + Required Side Clearance

Internal Booth Height $\geq$ Product Height + Required Top Clearance

Clearances must be defined by VITECH engineering rules and the actual product/loading method; the AI must not invent clearance values.

Step 4 – Calculate airflow

$$Q \text{ (m}^3/\text{s)} = A \text{ (m}^2) \times V \text{ (m/s)}$$

$$Q \text{ (m}^3/\text{h)} = A \times V \times 3600$$

$$Q \text{ (CFM)} = A \times V \times 2118.88$$

$$\text{CFM} = \text{CMH} \times 0.588578$$

$$\text{CMH} = \text{CFM} \times 1.69901$$

Use the **effective air-opening area** appropriate to the selected booth configuration. Do not automatically use gross booth dimensions if filters, baffles, product or structural members reduce the effective opening.

Step 5 – Select filtration

$$\text{Required Filter Area (m}^2) = \text{Airflow (m}^3/\text{s)} / \text{Design Filter Face Velocity (m/s)}$$

For CFM-based calculations, convert CFM to m³/s before applying a velocity in m/s. Filter velocity and allowable loading must be VITECH/filter-manufacturer approved values.

Step 6 – Calculate total pressure

$$\text{Total Pressure Requirement} = \text{Booth Loss} + \text{Filter Loss} + \text{Duct Loss} + \text{Fittings Loss} + \text{Damper Loss} + \text{Stack Loss} + \text{Other System Losses}$$

$$\text{Fan/Blower Selection Point} = \text{Required Airflow at Required Total/Static Pressure}$$

Step 7 – Select blower and motor

$$P_{\text{shaft}} \text{ (W)} = Q \text{ (m}^3/\text{s)} \times \Delta P \text{ (Pa)} / \eta_{\text{total}}$$

$$\text{Motor kW} = P_{\text{shaft}} / \text{Motor Efficiency}$$

$$1 \text{ HP} = 0.746 \text{ kW}$$

Final blower selection must be from the approved manufacturer fan curve at the calculated duty point. Never select a blower from CFM alone.

5. AIRFLOW & UNIT CONVERSION MASTER

Quantity	Conversion
1 m³/s	2118.88 CFM
1 m³/s	3600 CMH
1 CFM	1.69901 CMH
1 CMH	0.588578 CFM
1 Pa	0.10197 mmWG (approx.)
1 mmWG	9.80665 Pa
1 HP	0.746 kW
1 kW	1.34102 HP

6. PAINT BOOTH AIRFLOW – WORKED EXAMPLE

Example only. Replace the velocity with the approved VITECH value before using in production.

Suppose effective opening = 3.0 m × 2.5 m and assumed design velocity = 0.50 m/s.

$$\text{Area} = 3.0 \times 2.5 = 7.50 \text{ m}^2$$

$$Q = 7.50 \times 0.50 = 3.75 \text{ m}^3/\text{s}$$

$$\text{CMH} = 3.75 \times 3600 = 13,500 \text{ CMH}$$

$$\text{CFM} = 3.75 \times 2118.88 \approx 7,946 \text{ CFM}$$

Therefore the preliminary airflow duty is approximately 13,500 CMH / 7,950 CFM. This is an example to demonstrate the AI calculation chain, not a VITECH standard.

AI output should store: input values → formula → intermediate values → final result → unit → assumption/source → validation status.

7. FILTER SIZING

$$\text{Filter Area} = Q / V_{\text{filter}}$$

$$\text{Filter Quantity} = \text{Required Total Area} / \text{Area per Filter}$$

$$\text{Actual Filter Area} = \text{Quantity} \times \text{Individual Filter Area}$$

$$\text{Filter Loading} = \text{Airflow} / \text{Actual Filter Area}$$

The AI should compare calculated filter loading with the approved VITECH/filter supplier limit and flag any mismatch.

Filter database fields: filter type, media, dimensions, individual area, quantity, temperature rating, efficiency, pressure drop at design flow, maximum allowable loading, supplier/model and replacement reference.

8. DUCT DESIGN

Duct design should be calculated branch-by-branch and then main-line-by-main-line.

$$Q = A \times V$$

$$A \text{ (m}^2\text{)} = Q \text{ (m}^3/\text{s)} / V \text{ (m/s)}$$

$$\text{For a circular duct: } D = \sqrt{(4A/\pi)}$$

$$\text{For a rectangular duct: } A = \text{Width} \times \text{Height}$$

$$\text{Hydraulic Diameter, } D_h = 4A/P_{\text{wetted}}$$

The selected transport/design velocity must be stored by application and contaminant type. VITECH-specific values must be supplied to the AI.

Pressure loss should include straight duct friction, elbows, transitions, branches, dampers, filters, hoods, booth internals and stack/discharge losses.

9. BLOWER / FAN SELECTION

- Input: required CFM/CMH, total/static pressure, air temperature, density, altitude if relevant, duty type and operating hours.
- Select from the approved fan/blower manufacturer's performance curve.
- Check fan operating point, efficiency, RPM, impeller diameter, motor power, noise, rotation and stable operating region.
- Check motor service factor and electrical supply.
- Record selected model, duty point, pressure, flow, RPM, efficiency and motor rating in the equipment database.

$$\text{Air Power (W)} = Q \text{ (m}^3\text{/s)} \times \Delta P \text{ (Pa)}$$

$$\text{Shaft Power (W)} = \text{Air Power} / \text{Fan Efficiency}$$

$$\text{Motor Input Power} = \text{Shaft Power} / \text{Motor Efficiency}$$

10. PAINT BOOTH TYPES – DECISION MATRIX

Requirement	Preferred configuration to evaluate	Key checks
General wet painting	Cross Draft / Side Draft	Product size, airflow, filtration, layout
Large/heavy component	Downdraft / Cross Draft	Floor capacity, extraction, product handling
High containment	Closed booth	Door/air balance, leakage, exhaust
Open loading	Open Front	Capture velocity, enclosure effectiveness
High overspray / wet capture	Wet / Water Curtain	Water flow, pump, sludge handling, corrosion
Powder coating	Powder Booth	Recovery, filtration, powder loading
Manual powder	Manual Booth	Opening, operator access, recovery
Automatic powder	Automatic Booth	Gun arrangement, reciprocator, recovery

11. WET PAINT BOOTH / WATER CURTAIN CALCULATIONS

- Airflow: calculate from effective opening/face area and approved design velocity.
- Water circulation: determine from approved VITECH water loading / curtain design rule.
- Pump duty: flow + total dynamic head.
- Tank volume: working volume + required freeboard + sludge/maintenance allowance.
- Pump power: $P = \rho gQH/\eta$.
- Check water curtain uniformity, baffle arrangement, sludge removal, corrosion resistance and access.

$$\text{Pump Hydraulic Power (W)} = \rho \times g \times Q \times H$$

where ρ = liquid density (kg/m³), g = 9.81 m/s², Q = m³/s, H = m head.

VITECH-specific water flow per metre, pump selection and scrubber/booth rules must be encoded from approved historical calculations.

12. POWDER COATING BOOTH & RECOVERY

- Inputs: component size, production rate, powder consumption, gun quantity, manual/automatic operation, colour-change frequency, booth opening and recovery requirement.
- Calculate booth airflow and filter area.
- Calculate recovery airflow and pressure drop.
- Select cyclone and/or cartridge recovery according to powder loading and required recovery performance.

- Store cyclone dimensions, inlet/outlet, pressure drop, collection bin/hopper, blower and motor as linked components.
- Store cartridge number, filter area, media, pulse cleaning and compressed-air requirements.

$$\text{Powder Recovery Efficiency (\%)} = (\text{Recovered Powder Mass} / \text{Powder Entering Recovery System}) \times 100$$

Use measured/approved VITECH efficiency data for guarantees. Do not convert one historical efficiency into a universal design guarantee.

13. OVEN – HEAT LOAD FRAMEWORK

- Product heat load.
- Fixture/hanger/conveyor heat load.
- Air circulation/infiltration load.
- Wall/roof/floor conduction losses.
- Door opening losses.
- Exhaust losses.
- Required temperature and cure/dry time.
- Fuel/electrical heating method.

$$\text{Product Heat} = m \times C_p \times \Delta T$$

$$\text{Heat Duty} = \text{Product Heat} / \text{Heating Time}$$

$$\text{Total Heat Load} = \text{Product} + \text{Fixtures} + \text{Walls} + \text{Infiltration} + \text{Exhaust} + \text{Other Losses}$$

$$\text{Required Burner Input} = \text{Total Heat Load} / \text{Burner Efficiency}$$

$$\text{Electrical Heater kW} \approx \text{Total Heat Load (kW)} / \text{Heater Efficiency}$$

The AI must use the approved VITECH oven air-change/exhaust rule and actual burner/heater efficiencies. Do not assume these values.

14. DUST COLLECTOR – DESIGN FRAMEWORK

- Define dust type, particle size, concentration, temperature, moisture, explosibility/corrosiveness and required emission.
- Determine hood capture requirement and system airflow.
- Size ductwork and calculate total system pressure drop.
- Select cyclone, bag filter or cartridge collector.
- Calculate filter area from approved filtration velocity/air-to-cloth ratio.
- Size hopper, discharge valve/screw/rotary airlock where applicable.
- Select pulse valves, solenoids, sequence controller and compressed-air system.
- Select blower from duty-point curve.

$$\text{Filter Area} = \text{Airflow} / \text{Allowable Filtration Velocity}$$

$$\text{Air-to-Cloth Ratio} = \text{Airflow} / \text{Filter Area}$$

The allowable filtration velocity must be stored by dust type/filter media and approved source.

15. SCRUBBER DESIGN FRAMEWORK

- Identify contaminant and required removal efficiency.
- Determine gas flow, temperature and composition.
- Select spray tower, packed bed, orifice, horizontal wet scrubber or activated-carbon system.
- Calculate vessel diameter/height from approved gas velocity/design criteria.
- Calculate liquid circulation, L/G ratio, nozzle flow/pressure and pump duty.
- Calculate pressure drop and select blower.
- Specify MOC based on chemical compatibility and temperature.

$L/G \text{ Ratio} = \text{Liquid Flow Rate} / \text{Gas Flow Rate}$

$\text{Pump Power} = \rho g Q H / \eta$

VITECH-specific scrubber sizing rules, including horizontal-baffle height rules, must be stored as explicit selection rules.

16. CONVEYOR CALCULATIONS

$\text{Production Rate} \approx \text{Conveyor Speed} / \text{Hook Pitch}$ (with consistent units)

$\text{Total Load per Hook} = \text{Product} + \text{Fixture} + \text{Hook/Carrier}$

$\text{Total Moving Load} = \text{Number of Loaded Hooks} \times \text{Load per Hook}$

Drive Power depends on chain pull, friction, elevation, bends, acceleration and drive efficiency.

The AI must retain actual VITECH conveyor design rules for chain size, hook pitch, support spacing, drive selection and allowable load.

17. MATERIAL MASTER

- Material name and grade.
- Thickness/range.
- Application.
- Temperature limit.
- Corrosion resistance.
- Chemical compatibility.
- Fabrication method.
- Surface finish/coating.
- Approved supplier/standard.
- VITECH internal approval status.

Do not let the AI substitute SS/PP/FRP/MS/GI automatically unless a VITECH rule or engineering requirement authorizes the substitution.

18. COMPONENT MASTER

- Blower/fan: manufacturer, model, CFM, pressure, RPM, efficiency, impeller, motor and curve reference.
- Motor: make, model, kW/HP, RPM, voltage, phase, frequency, efficiency, IP class, mounting.
- Filter: type, media, size, area, rating, temperature and pressure-drop curve.
- Pump: make/model, flow, head, power, RPM, MOC and seal.
- Valve/damper: type, size, pressure, MOC and actuation.
- Pulse system: valve, solenoid, controller, manifold and compressed-air requirement.
- Instrumentation: pressure gauge, manometer, temperature, flow, sensor and control device.

19. BOM STRUCTURE

System BOM → Subassembly BOM → Manufacturing BOM → Purchase/Bought-Out BOM

- System BOM: complete equipment scope.
- Subassembly BOM: blower assembly, filter assembly, pump assembly, recovery assembly etc.
- Manufacturing BOM: sheets, folds, stiffeners, frames, supports, fasteners and fabrication parts.
- Purchase BOM: motors, blowers, filters, pumps, valves, electrical items and bought-out components.
- Every BOM line must have part number, description, specification, material, quantity, unit, make/model where applicable and source drawing.

20. GA / DRAWING DATABASE

- Overall L×W×H and datum references.
- Booth opening, doors, filters, plenum, blower, motor, duct and stack positions.
- Maintenance clearances.
- Foundation/support points.
- Lighting and electrical locations.
- Filter-bank setting-out.
- Component setting-out rules.
- Nozzle, pump, tank, duct take-off and access-door locations.
- Drawing number, revision, project/customer and approval status.

21. TECHNICAL SPECIFICATION GENERATION

For each selected equipment item the AI should generate a controlled specification containing:

- Scope of supply and exclusions.
- Equipment type, application and capacity.
- Dimensions and MOC.
- Airflow/CMH/CFM and velocity.
- Pressure/static pressure and pressure-drop basis.
- Filter details.
- Blower/fan and motor details.
- Pump/heating/recovery/conveyor details where applicable.
- Ducting, stack and accessories.
- Electrical/control requirements.
- Safety and instrumentation.
- Testing, inspection, performance requirements and documentation.
- Drawing/BOM/specification revision references.

22. AI VALIDATION RULES – MUST BE BUILT IN

- Never produce a final design if a critical customer input is missing.
- Separate ASSUMED, CUSTOMER-CONFIRMED and VITECH-STANDARD values.
- Every formula result must retain input values, units and source.
- Every selected bought-out component must be checked against the calculated duty point.
- Do not use a historical project as a universal standard unless the database marks it as a reusable rule.
- Flag values outside approved VITECH ranges.
- Flag contradictory inputs, such as booth dimensions smaller than product plus required clearance.
- Flag insufficient filter area, excessive pressure drop, motor overload and inadequate duct velocity.
- Flag missing electrical supply, installation condition, safety requirement or emission requirement.
- Final manufacturing release requires engineering approval.

23. REQUIRED AI ANSWER FORMAT

When a customer asks for a paint booth, the AI should answer in this sequence:

1. Requirement Summary 2. Missing Inputs 3. Recommended Booth Type 4. Reason for Selection 5. Design Dimensions 6. Airflow Calculation (CMH + CFM) 7. Filter Selection 8. Duct Calculation 9. Pressure Drop 10. Blower Selection 11. Motor Selection 12. Structure/Panel Concept 13. BOM 14. GA Inputs 15. Technical Specification 16. Assumptions 17. Validation Checks 18. Approval Required

The AI should not simply output a quotation. It should show the engineering basis behind the quotation.

24. EXAMPLE AI ENGINEERING RECORD

Customer: Industrial wet painting. **Product:** 2500 × 1500 × 1800 mm. **Process:** Manual wet paint. **Preliminary booth:** Cross Draft, subject to confirmation.

Step A: Determine internal booth dimensions from product + approved clearance rules.

Step B: Determine effective opening area.

Step C: $Q = A \times V$.

Step D: Convert Q to CMH and CFM.

Step E: Determine filter area and quantity.

Step F: Calculate booth/filter/duct/fitting/stack pressure losses.

Step G: Select blower at calculated Q and pressure.

Step H: Select motor from blower duty and efficiency.

Step I: Generate parametric structure/panel dimensions.

Step J: Generate BOM and technical specification.

This record must retain the exact source and approval status of every assumed design value.

25. DATA SOURCE / TRAINING PRIORITY

Priority	Source to provide Spark	AI value
1	Actual VITECH calculation Excel with formulas	Highest – captures engineering logic
2	Issued GA/DWG/DXF	High – captures geometry and setting-out
3	Issued BOM Excel	High – enables automatic BOM
4	Issued quotations	High – quotation/specification mapping
5	Technical specifications	High – controlled output language
6	Historical project package	Very high – enquiry to final design chain
7	Product brochures/catalogues	Medium – product identification
8	Standards/catalogues	Reference – external engineering basis

26. CRITICAL VITECH DATA THAT MUST STILL BE CONFIRMED

- Face velocity by paint-booth type.
- Default air-change rate/exhaust rule for drying rooms and ovens.
- Complete costed paint-booth BOM.
- VITECH component setting-out rules.
- Horizontal wet-scrubber height/design rule.
- Powder-coating plant calculations.
- Pollution-control equipment calculation sheets.
- More water-wash booth historical designs.
- More powder-coating plant historical designs.
- LPG-fired oven calculations/designs.
- Standard margins and bought-out item mark-up rules.

- Approved standard dimensions, materials, sections, filter sizes and equipment models.

These are critical because the AI should distinguish **known VITECH engineering rules** from **generic engineering assumptions**.

27. FINAL DATA MODEL FOR SPARK INVENTIVE

PRODUCT → TYPE → MODEL → APPLICATION → INPUTS → ASSUMPTIONS → FORMULA → CALCULATION → OUTPUT →
SELECTION RULE → COMPONENT → BOM → COST → GA → P&ID; → SPECIFICATION → VALIDATION → QUOTATION →
PROJECT HISTORY → REVISION

Recommended implementation: maintain the master data in structured Excel/CSV/database tables, while keeping native calculation workbooks, CAD drawings and issued documents linked by document ID and revision. The AI should retrieve the source document and not learn only the final PDF text.

28. ENGINEERING DISCLAIMER FOR AI PLATFORM

This knowledge base is an engineering framework for training and structuring the VITECH AI platform. It is not a substitute for approved VITECH engineering drawings, calculations, standards, customer confirmation, risk assessment or final design approval. Generic formulas must not be converted into VITECH-specific design constants without source approval. Final manufacturing release must remain subject to authorized VITECH engineering review.